Claude-powered procurement intelligence.

An evidence-oriented comparison workspace: React + Express + PostgreSQL for a durable procurement workflow; deterministic TypeScript for financial results; and an official Anthropic SDK integration for structured extraction, tool-assisted analysis and negotiation proposals.

VERIFIED LOCAL WORKFLOW

20 tests with PostgreSQL | upload, comparison, scenario, decision and restart checks

LIVE CLAUDE NOT VERIFIED: Anthropic API key absent at 2 October 2026 checkpoint.

01 / APPLICATION BOUNDARIES

REACT	EXPRESS	DOMAIN	POSTGRESQL
Supplier workspace	Routes / validation	Comparison / audit	Facts / revisions

Claude API stays behind the server and receives bounded, event-scoped evidence and computed results.

SOURCE PIPELINE

Contract provenance

INGEST

Text-bearing PDF, DOCX and UTF-8 TXT; 10 MB maximum.

LOCATE

Chunks retain document, character/line locations and excerpts.

VALIDATE

Reject absent excerpts, invalid source IDs and impossible figures.

VERSION

Superseded facts stay available for old evidence references.

CALCULATION BOUNDARY

Financial authority

PROJECT

Annual price compounds at the evidenced escalation rate.

COMPARE

Total = rounded annual recurring costs + setup fee.

SCORE

Six weighted criteria: cost, flexibility, SLA, liability, warranty, support.

FAIL CLOSED

Missing weighted evidence makes overall score unavailable.

DESIGN RULE A supplier ranking is not authority to sign. Claude cannot overwrite the deterministic engine or create a procurement approval.

Trust the evidence. Keep the decision human.

Source-linked risk interpretation and proposed negotiation targets operate inside an auditable, revision-aware workflow. The locally executed checkpoint demonstrates persistence and safety boundaries, not completed live-model acceptance.

02 / ANTHROPIC INTEGRATION DESIGN

Official Anthropic SDK

Claude Sonnet 4.6 is the configured default, using JSON Schema structured output validated by Zod. The Messages/tool loop is bounded to eight rounds, with a 120-second request timeout.

Four event-scoped tools

get_supplier_profile, get_procurement_requirements, get_contract_evidence, and calculate_projected_cost. Tool arguments and returned IDs are validated against the active event.

03 / STATE, DECISIONS AND SECURITY

Transactional revisions

Editing terms, suppliers or requirements increments the event revision. Stale analyses are labelled and cannot be accepted.

Preserved evidence

PostgreSQL stores versioned analyses, risks, opportunities, term references, human decisions and audit events.

Single-user boundary

Loopback-only development. There is no login, tenancy or role-based permission layer; internet deployment is unsafe.

Sensitive data

Original uploads are stored locally without encryption. Actual Claude requests would send selected document text to Anthropic.

04 / VERIFIED CHECKPOINT

LOCAL CHECKS PASSED

Build, migrations, idempotent seed, 20 tests, isolated API exercises, file validation, transaction/version behavior, scenario recalculation and persistence across API restarts.

NOT YET VERIFIED

Real Claude extraction and tool-assisted reasoning, fresh browser acceptance, public deployment, multi-user security, OCR and legal correctness.

REPRODUCIBLE WHAT-IF: SUPPLIER B

8% to 5%

R716,036.39 projected five-year difference

A hypothetical change, not an agreed supplier concession.

IMPLEMENTATION SOURCE

github.com/keenosmith-del/claude-powered-legal-procurement-intelligence